Q1 b)

-0.75

Integral part: 0 = 0

Fractional part: 0.75 = 0.11

0.75 x 2 = 1.5 -> 1

0.5 x 2 = 1.0 -> 1

Normalize: 0.75 = 0.11 = 1.1 x 2^-1

Sign bit = 1

Bias = 2^(8-1) -1 = 127

Exponent = bias + (-1) = 127-1=126 = 0111 1110

Fraction = 1000… 0000 (22 0s after 1)

Final bit pattern (32 bits) = 1 01111110 1000…0000

Its hex = BF400000

Q2 b)

BF900000 = 1011 1111 1001 00…00 (20 0s after 1001)

= 1 0111 1111 00100…00

Sign bit = 1, a negative number

Exponent = 8 bits = 0111 1111 = 127

Convert exponent from bias 127 notation: 127-127=0

Fraction = 23 bits = 001

So the number in binary is 1.001

1.01 = 1.125

BF900000 => -1.125

Q3 b)

9 ^ 5

9: 1001

5: 0101

^: 1100 = C = 12

Q4 a)

C3 | 5F

C3 : 1100 0011

5F : 0101 1111

OR: 1101 1111 = DF = 223

Q4 b)

DE << 3

DE : 1101 1110 << 3 = 1111 0000 = F0 = 240

Q5 a)&b) check Workshop3Qs567.cpp

12 >> 2 = 00..00 1100 (28 0s ahead 1100) >> 2 = 00..00 0011 = 3

1 << 2 = 00..00 0001 (28 0s ahead 0001) << 2 = 00..00 0100 = 4

Q6 a) check Workshop3Qs567.cpp

C= 0xA4 = 1010 0100 >> 2 = 1110 1001

Two's complement

1110 1001 =>

0001 0110 + 1 = 0001 0111 = 23

1110 1001 = -23

Q6 b)

C = 1010 0100 >> 2 = 0010 1001 = 41

Q7 a)

C = 0xA4 = 1010 0100

Mask = 0000 0001

  1010 0100

& 0000 0001

  0000 0000

Q7 c) check Workshop3Qs567.cpp

C=0xA4 = 1010 0100

Mask: 0000 0111 = 0x07

& : 0000 0100

Q7 d) check Workshop3Qs567.cpp

C=0xA4 = 1010 0100

Mask: 1110 0000 = 0xE0

& : 1010 0000